

Trainings ▾

Install

with Mechanically Actuated Injector

Do **not** lubricate the o-rings until the injector is ready for installation in the cylinder head.

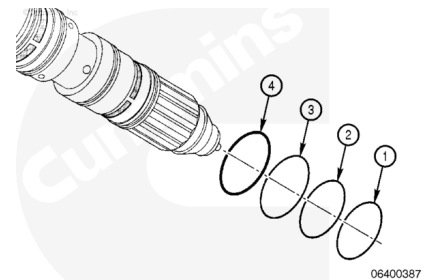
Install four new o-rings on the injector. Do **not** twist the o-rings.

O-ring (1) is brown and is smaller in diameter than the other three.

O-ring (2) is green.

O-ring (3) is light blue and is larger in diameter than (1), but has a smaller cross section than o-ring (4).

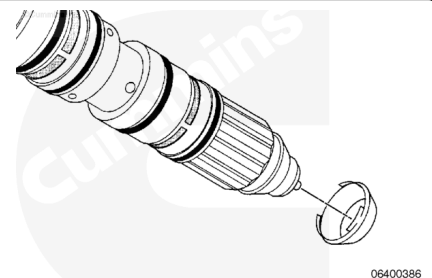
O-ring (4) is black and is the largest in diameter and cross section.



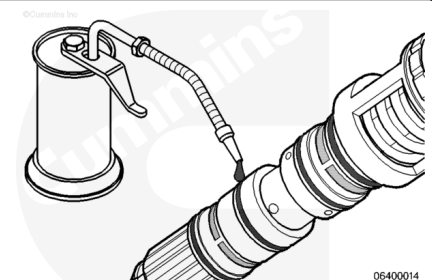
Install a new, proper thickness sealing ring which corresponds to the cylinder location noted during removal.

Note : If installing an injector for the first time into a resurfaced cylinder head, use the following procedure to determine proper sealing ring thickness. Refer to Procedure 002-022 in Section 2.

(/qs3/pubsys2/xml/en/procedures/20/20-002-022.html)



Use clean lubricating engine oil (15W-40) to lubricate the o-rings.



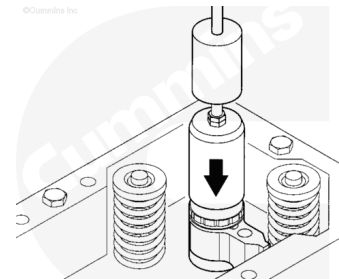
Install the hold-down clamp on the side of the injector.

Install the injector and hold-down clamp in the cylinder head.

Note : If the injector installation tool, Part Number 3824830, is not available, use a 40 mm [1 9/16 in] socket with an extension and a rubber mallet to install the injector.

Use injector installation tool, Part Number 3824830, to set the injector in the bore.

The slide hammer will make a dull sound when the injector is seated properly.



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Use clean engine oil to lubricate the injector hold-down capscrew.

Install and tighten the injector hold-down capscrew.

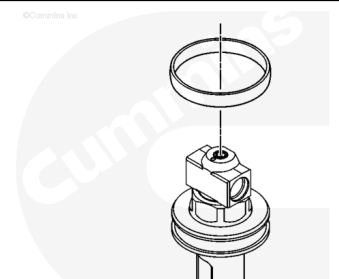
Torque Value: 54 n•m [40 ft-lb]



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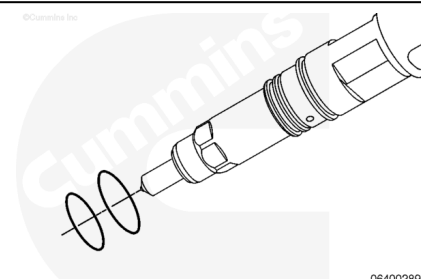
with Electronically Actuated Injector

Install the new o-ring/damper on the injector.



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Install the new o-rings on the injector body.

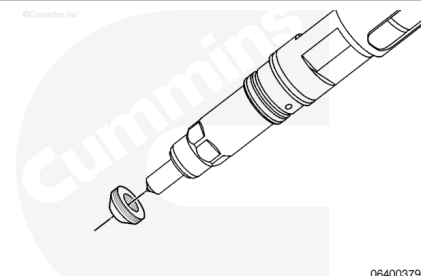


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⚠ CAUTION ⚠

Do not reuse the injector hold-down capscrew or injector combustion seal. The reuse of those components can result in engine damage.

Install the new seal on the end of the injector.



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⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

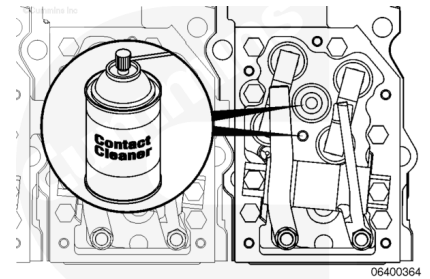
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Keep the injector bore and combustion chamber protected when using compressed air to prevent contamination. Failure to do so can cause engine damage.

⚠ CAUTION ⚠

Do not reuse the injector hold-down capscrew or injector combustion seal. The reuse of those components can result in engine damage.

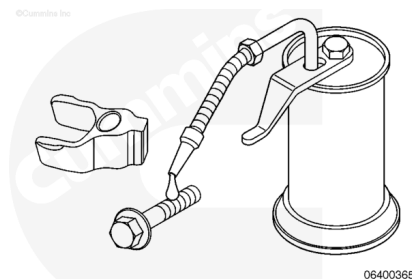


Note : On applications with high drain line head pressure, such as a fuel tank mounted higher than the engine, it is possible for fuel to leak from the drain line into the cylinder when the injector is removed. Prior to installing the injector, check for fuel dripping into the cylinder and evacuate the fuel before installing the injector.

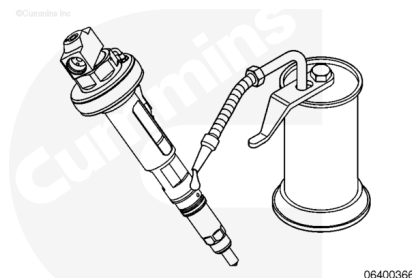
Clean the injector bore in the cylinder head with electrical contact cleaner, Part Number 3824510, or equivalent, to make sure the o-rings will seal properly.

Clean the threaded hole in the cylinder head where the injector hold-down capscrew will be installed with electrical contact cleaner, Part Number 3824510, or equivalent, and compressed air.

Lubricate the threads and underside of the new injector hold-down capscrew with clean lubricating engine oil.

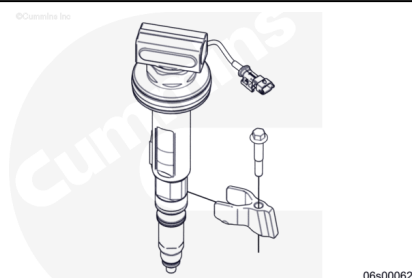


Lubricate all of the o-rings with a light coating of clean lubricating engine oil.



⚠ CAUTION ⚠

Do not reuse the injector hold-down capscrew or injector combustion seal. The reuse of those components can result in engine damage.



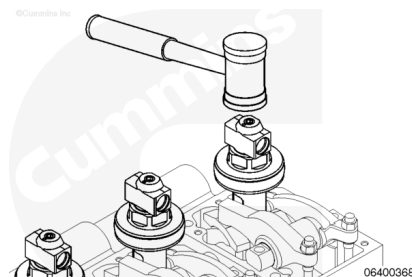
Assemble the injector hold-down clamp and capscrew into the injector.

⚠ CAUTION ⚠

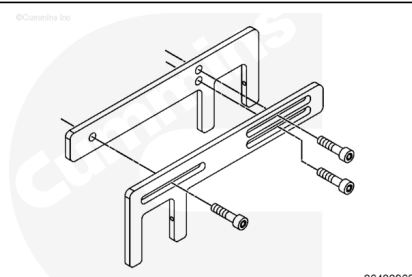
Only use a soft face mallet. Otherwise damage to the injector can occur.

The injector installer can be used on single or multiple injector installations.

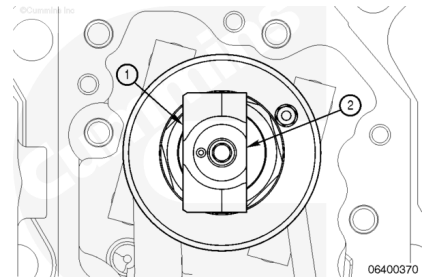
Install the assembly into the cylinder head by tapping gently on the injector with a soft face mallet.



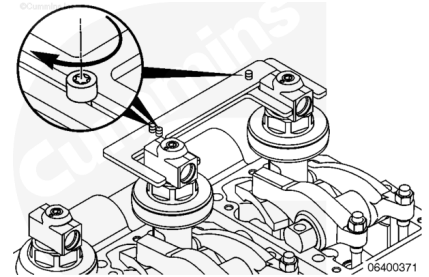
Assemble the injector installer kit, Part Number 4918620. Hand tighten the three shoulder capscrews.



Place the injector installer on the injector such that the ball spring plunger is on the tapered side of the injector (1). If necessary, adjust the ball spring plunger so that it presses against the injector. Make sure the flat side of the injector (2) sits flush with the injector installer.



Align the injector installer with two injectors.



Tighten the injector hold-down capscrews.

The 60 degree torque angle can be achieved at the capscrew by marking the location of one corner of the capscrew to a location on the injector hold-down clamp.

Tighten to:

Torque Value: 68 n•m [50 ft-lb]

Back off completely.

Tighten to:

Torque Value: 30 n•m [22 ft-lb]

Rotate the capscrew 60 degrees.

Remove the injector installer.

